

Assignment 2

NIS3352-01 Modern Cryptography(2)

Marking Rubrics for Assignments.

- You will receive full marks if your idea is (mostly) correct and you haven't showed an adequate effort to present your ideas. No marks will be deducted if there are small errors or typos.
- If you are missing an important point of a problem, yet you have partially answered it, 10 marks of the question will be deducted.
- Full marks of the problem will be deducted if (1) your idea is not correct at all or (2) there is an obvious sign that you are copying it from your classmates.
- Each assignment is given one week to finish. Yet, we still accept it if it is at most one week late. For assignments two weeks after the deadline, 10 marks will be deducted.

Problem 1 (25 marks) For the Schnorr signature scheme, if Alice reuses the same random number r to sign two distinct messages M_1, M_2 whose corresponding signatures are σ_1, σ_2 . Is it secure? Explain the reason.

Problem 2 (25 marks) Consider a variant of DSA in which H is omitted. (So the computation of $e = H(M)$ is omitted and $z = r^{-1} \cdot (M + ds)$.) Show that this variant is not secure.

Problem 3 (25 marks) Show that the DSA identification scheme is UI-PA secure if the Discrete Logarithm problem is hard.

Here is how the DSA identification protocol works. Let G be a cyclic group of prime order p with generator $g \in G$. The DSA identification protocol is $\mathcal{I}_{\text{DSA}} = (G, P, V)$, where:

- The key generation algorithm $\mathcal{G}$ runs as follows:

$$s \xleftarrow{R} \mathbb{Z}_p, \quad h \leftarrow g^s$$

The verification key is $vk := h$, and the secret key is $sk := s$.

- The protocol between P and V runs as follows:

1. P computes $r \xleftarrow{R} \mathbb{Z}_p$, $R := g^r \in G$, and sends R to V .
2. V computes $e \xleftarrow{R} \mathbb{Z}_p$, $d \xleftarrow{R} \mathbb{Z}_p$, and sends (e, d) to P .
3. P computes

$$z := r^{-1} \cdot (e + d \cdot s) \in \mathbb{Z}_p$$

and sends z to V .

4. V checks if

$$R = (g^e \cdot h^d)^{z^{-1}};$$

if so V outputs accept; otherwise, V outputs reject.

Problem 4 (25 marks) Show that

DSA identification scheme is UI-PA secure + H, F are random oracles $\implies$ DSA signature scheme is EUF-CMA secure.